

FAIRHOPE, AL, USA

WMO: 720592

Lat: 30.460N

Lon: 87.877W

Elev: 92

StdP: 14.65

Time Zone: -6.00 (NAC)

Period: 08-19

WBAN: 00186

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	29.6	32.1	14.4	11.4	36.7	19.2	14.5	39.1	21.1	66.6	19.2	62.0	4.1	0	0.350

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.7	93.4	78.6	91.1	78.1	90.1	77.8	81.9	87.5	81.2	86.3	80.5	85.6	6.3	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.1	162.0	83.8	79.4	153.3	83.2	79.1	151.6	83.1	45.6	86.8	44.7	85.9	44.0	85.8	86.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
18.1	16.0	13.3	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	67.4	51.3	55.7	60.3	66.6	73.0	80.0	81.1	80.8	77.8	68.8	58.0	54.6
	DBStd	12.62	9.92	9.21	7.98	5.66	5.02	3.77	3.10	3.29	3.66	6.95	8.17	8.90
	HDD50	241	106	45	14	0	0	0	0	0	0	0	22	54
	HDD65	1570	436	277	185	47	8	0	0	0	0	43	238	336
	CDD50	6589	145	206	333	498	714	900	964	955	834	582	261	197
	CDD65	2443	11	18	39	95	257	450	499	490	384	160	26	14
	CDH74	22560	15	36	130	651	2165	4740	5249	4900	3162	1364	130	18
	CDH80	8082	0	0	7	68	536	1903	2103	1976	1151	333	5	0
Wind	WSAvg	5.0	6.3	6.5	6.5	6.1	4.9	4.1	3.3	3.3	3.6	4.7	4.8	6.0
Precipitation	PrecAvg	62.2	5.1	4.6	5.4	4.7	4.2	5.8	7.4	6.3	5.9	3.6	4.4	4.8
	PrecMax	86.5	15.8	9.8	11.6	17.0	13.3	13.9	15.7	10.0	22.8	12.6	11.7	12.8
	PrecMin	41.3	1.2	1.1	0.6	0.8	0.7	1.3	2.3	2.4	1.6	0.0	1.2	1.9
	PrecStd	11.6	3.2	2.1	3.0	3.4	2.8	3.2	2.9	2.4	4.2	3.1	3.0	2.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	76.5	77.4	81.2	84.0	90.4	96.6	96.6	96.8	92.8	88.4	81.1	76.9
		MCWB	69.4	68.9	67.3	72.2	74.6	77.4	79.5	78.9	77.2	75.7	70.4	68.6
	2%	DB	72.6	73.5	77.4	81.7	87.6	92.9	92.8	93.2	90.4	86.1	77.4	73.1
		MCWB	68.5	68.6	68.3	70.0	74.3	78.0	78.9	79.6	77.1	73.9	68.8	68.8
	5%	DB	70.4	72.2	74.6	80.7	85.6	90.6	90.6	90.4	88.3	83.6	74.7	72.0
		MCWB	67.8	67.4	67.5	69.5	73.8	77.3	78.7	78.8	76.5	74.2	67.3	68.5
	10%	DB	68.0	70.0	72.4	78.6	82.2	88.4	88.4	88.3	86.3	81.5	72.3	69.8
		MCWB	65.4	66.5	66.9	68.8	73.0	77.1	78.7	78.3	76.5	71.8	65.9	67.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	71.9	72.6	72.4	76.6	78.5	82.2	82.8	83.3	81.3	81.2	74.3	72.5
		MCDB	74.2	74.9	75.8	80.4	81.1	87.8	88.4	91.4	86.9	84.4	77.4	74.0
	2%	WB	70.0	70.6	71.3	74.4	77.2	81.2	81.7	82.1	80.1	78.8	72.4	70.5
		MCDB	71.8	72.5	74.5	78.5	81.9	86.1	86.9	88.0	85.3	82.4	75.7	71.7
	5%	WB	68.3	69.3	69.9	73.2	76.1	80.3	81.0	81.3	79.1	76.6	70.1	69.3
		MCDB	69.9	70.7	72.5	77.2	82.0	85.5	86.3	86.2	84.2	80.9	72.8	70.5
	10%	WB	65.4	67.0	68.6	71.8	75.0	79.3	80.3	80.6	78.1	74.4	67.7	67.2
		MCDB	68.1	68.8	71.3	76.2	81.3	84.7	85.6	85.4	83.4	79.4	70.8	68.9
Mean Daily Temperature Range	5% DB	MDBR	19.7	18.7	19.3	19.7	18.2	15.3	14.7	15.2	16.9	21.4	22.0	18.2
		MCDBR	17.7	17.6	18.6	21.6	20.7	18.3	17.1	17.5	18.9	21.7	21.8	17.9
		MCWBR	13.8	12.1	11.1	11.8	9.4	6.9	7.0	7.4	8.3	11.0	14.2	14.1
	5% WB	MCDBR	14.8	14.2	14.6	15.3	16.0	14.2	14.4	14.2	15.4	15.0	17.8	15.7
		MCWBR	13.0	11.4	10.6	10.8	8.5	7.5	7.4	7.7	8.3	9.6	14.1	14.4
Clear-Sky Solar Irradiance	taub		0.332	0.349	0.373	0.403	0.438	0.463	0.509	0.498	0.444	0.383	0.355	0.341
	taud		2.547	2.503	2.430	2.347	2.285	2.267	2.160	2.185	2.303	2.458	2.477	2.531
	Ebn at Noon		286	290	289	283	272	264	251	253	263	274	274	276
	Edh at Noon		27	31	36	40	43	43	48	46	40	32	29	26
All-Sky Solar Radiation	RadAvg		973	1155	1564	1896	2102	1996	1876	1757	1653	1449	1110	855
	RadStd		76	158	123	152	108	151	131	130	99	144	96	77

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (30 neighbors)	+0.85	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+288	+187

Nomenclature: See separate page